Delinquency Prediction Model

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System Workflow and Data Pipeline

The delinquency prediction system follows a structured end-to-end pipeline designed to identify customers at risk of missing loan or credit payments. The workflow begins with data collection from 500 customer records containing 19 features including demographics, financial metrics, and payment history across six months. Data quality issues were systematically addressed: 7.8% missing income values were imputed using employment-status-based means, credit utilization outliers exceeding 100% were clipped, and employment status inconsistencies were standardized. Feature engineering created derived variables such as age demographics (Young/Middle-Aged/Senior) and credit score ratings (Poor/Average/Good/Excellent), along with a continuous missed payment indicator identifying customers with four consecutive missed payments. The refined dataset was split 70-30 for training and testing, with SMOTE applied to address the 84-16 class imbalance, enabling more balanced model learning.

Pipeline Stage	Key Activities	Output
Data Collection	500 customer records, 19 features	Raw dataset with quality issues
Data Cleaning	Handle missing values, fix inconsistencies	0 duplicates, standardized categories
Feature Engineering	Create derived variables, payment patterns	19 final predictive features
Train-Test Split	70-30 stratified split with SMOTE	Balanced training and test sets
Model Training	Logistic Regression on SMOTE-resampled data	Trained classifier with coefficients
Evaluation	Accuracy, ROC-AUC, precision, recall metrics	Performance benchmarks for deployment

Agentic AI Roles and Decision Support Architecture

The delinquency prediction system operates as an intelligent decision-support agent rather than a fully autonomous decision-maker, with clearly defined roles and human oversight checkpoints. The Risk Assessment Agent continuously monitors incoming customer transactions and profile changes, scoring each customer's delinquency probability based on credit utilization, debt-to-income ratio, missed payment history, and employment status. The Alert and Escalation Agent flags high-risk customers (those with continuous missed payments or utilization above 50%) and routes them to appropriate intervention teams—payment reminder specialists for moderate risk, financial counseling for high risk, and credit review teams for critical cases. The Model Monitoring Agent tracks prediction accuracy and fairness metrics monthly, detecting concept drift when model performance degrades or when predictions become biased against specific customer segments. The Human Review Agent, staffed by bank loan officers and risk managers, makes final decisions on customer interventions, credit limit adjustments, or loan modifications. This layered approach ensures that AI identifies risk patterns while humans retain authority over consequential decisions, maintaining accountability and allowing context-aware judgment for edge cases or special circumstances.

Risk Assessment Agent

Scores delinquency probability using financial metrics and payment history; updates scores monthly as new data arrives

Alert and Escalation Agent

Routes flagged customers to intervention teams based on risk severity; sends proactive payment reminders and financial counseling offers

Model Monitoring Agent

Tracks accuracy, fairness, and concept drift; triggers retraining when performance drops below 60% recall on delinquent customers

Human Review Agent

Approves all customer interventions; overrides AI predictions when

Responsible AI Guardrails and Ethical Safeguards

The delinquency prediction system incorporates multiple guardrails to ensure fair, transparent, and accountable AI deployment. Fairness safeguards prevent discrimination by monitoring prediction rates across demographic segments and preventing location, age, or employment-status bias; the model explicitly avoids using protected characteristics as primary decision drivers. Explainability guardrails mandate that every high-risk customer flagged receives an explanation of top three factors driving their risk score—for example, "High delinquency risk due to: 65% credit utilization, 0.35 debt-to-income ratio, and two consecutive missed payments"—enabling customers and staff to understand and challenge predictions. Transparency requirements ensure that customers are informed when their account is being monitored by a predictive system and have the right to request a manual review. Data privacy protections limit model access to approved financial metrics only, encrypt all customer data at rest and in transit, and prevent unauthorized sharing of risk scores. Human-in-the-loop controls mandate that no automated action (credit limit reduction, account freeze, or collection referral) occurs without human approval, preserving customer dignity and allowing for nuanced judgment. Model drift detection continuously monitors for performance degradation and automatically triggers retraining when accuracy drops, preventing outdated predictions from causing harm. Regular audits assess model bias, accuracy, and fairness quarterly, with results reviewed by a cross-functional governance committee including risk, compliance, and ethics representatives.

Fairness Monitoring

Track prediction rates across age groups, employment status, and location; alert if any segment shows >10% deviation in delinquency flagging rates

Explainability Standards

Provide top-3 factor explanations for every flagged customer; enable customers to request human review of AI decisions

Data Minimization

Use only necessary financial features; exclude sensitive attributes like race, gender, or religion from model inputs

Human Approval Workflow

Require manager sign-off before sending collection notices or credit limit reductions based on AI predictions

Continuous Retraining

Retrain model every 3-6 months using recent data; compare new model performance to baseline before deployment

Audit and Governance

Conduct quarterly fairness, bias, and accuracy audits; maintain public-facing report on model performance and limitations

Business Impact and Risk Mitigation

The delinquency prediction model delivers measurable business value through early risk identification and targeted interventions that reduce loan losses and improve customer outcomes. With 500 customers analyzed, approximately 80 (16%) are currently delinquent; the model identified key risk factors that concentrate delinquency among specific segments—customers with 4+ consecutive missed payments show 25% delinquency rate versus 16% baseline, while unemployed customers with high credit utilization exceed 30% risk. By identifying these patterns, the bank can intervene early: for 50 high-risk customers flagged monthly, proactive payment reminders and financial counseling are expected to prevent delinquency in 5-8 customers annually, translating to approximately $80,000-$150,000 in avoided loan losses (assuming average $15,000-$20,000 loss per defaulted customer). The model's SMART objectives target reducing delinquent accounts by 10-15% within 12 months and improving recall of delinquent customer detection from current 29% to above 60% through model refinement. Beyond financial metrics, the system improves customer experience by offering early support to struggling borrowers rather than pursuing collections after default, strengthening bank-customer relationships and retention.

Business Metric	Current State	Target (12 Months)	Expected Impact
Delinquency Rate	16% (80 of 500)	10-12%	Reduce loan losses by $120K–$180K annually
Model Recall for Delinquent Customers	29%	Above 60%	Catch 3-4x more at-risk customers early
Intervention Success Rate	Baseline (pre-model)	15-20% prevented delinquency	Convert 5-8 flagged customers monthly to on-time payers
Customer Satisfaction	Not tracked	Improve by 10%	Early counseling reduces financial stress vs. collections
Operational Efficiency	Manual review only	60% automated flagging	Reduce analyst review time by ~8 hours per week